

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

Department of Computer Science Engineering

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Course Title: AI Assisted Coding

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Problem 1: AI-Assisted Bug Detection

Buggy Code

```
def factorial(n):  
    result = 1  
    for i in range(1, n):  
        result = result * i  
    return result
```

Corrected Code

```
def factorial(n):  
    if n < 0:  
        return "Invalid"  
    result = 1  
    for i in range(1, n + 1):  
        result *= i  
    return result
```

Explanation: Off-by-one error fixed and input validation added.

Problem 2: Improving Readability

Original Code

```
def calc(a, b, c):  
    if c == "add":  
        return a + b  
    elif c == "sub":  
        return a - b
```

Improved Code

```
def calculate(a, b, operation):  
    if operation == "add":  
        return a + b  
    elif operation == "sub":  
        return a - b  
    else:  
        return "Invalid"
```

Explanation: Improved naming and readability.

Problem 3: PEP8 Standards

Improved Code

```
def check_prime(n):  
    if n <= 1:  
        return False  
    for i in range(2, n):  
        if n % i == 0:  
            return False  
    return True
```

Explanation: Function follows Python naming conventions.

Problem 4: Code Review

Improved Code

```
def double_even_numbers(numbers):  
    if not isinstance(numbers, list):  
        return "Invalid input"  
    return [x * 2 for x in numbers if isinstance(x, int) and x % 2 == 0]
```

Explanation: Added validation and better naming.

Problem 5: Performance Optimization

Optimized Code

```
def sum_of_squares(numbers):  
    return sum(x * x for x in numbers)
```

Explanation: Using generator expressions improves performance.

Conclusion

AI-based code review helps identify bugs, improve readability, and optimize performance efficiently.

Human validation is essential for final correctness.